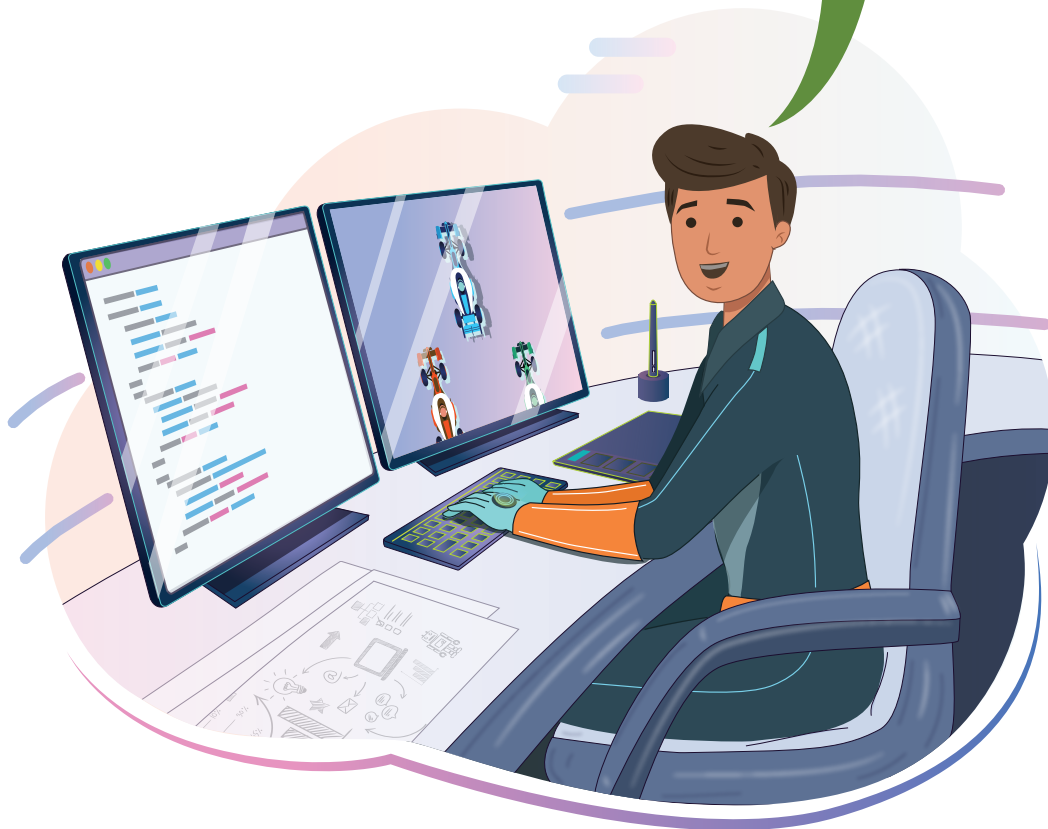




Game Design Process

I love playing video games, just like you! Most of the games I play are made by me. You can create and play your own games too!



Computer games have really gained popularity since the beginning of this century. Computers can generate virtual environments for a broad range of games, making it an immersive experience for users. We access online games through an active internet connection. Some have dedicated client systems and others just need a web browser.

Online games are of different genres. Role-playing games, shooters, first-person shooters, side-scrollers, and platformers are just a few.

A video game is an electronic game where a player interacts with objects and characters displayed on a screen. It can be played individually or in teams. We can access video games through CDs, DVDs, or digital downloads. Players use an input device like a joystick, controller, or keyboard to interact with the game. The Xbox One, PlayStation 4, and Wii U are some newer video game consoles. We can use PCs to play video games, which are also called PC games.



Why play video games?

- a. Video games help us develop negotiation and strategizing skills to reach a goal.
- b. Multi-player video games teach us to work in a team.
- c. Video game design helps us take better decisions, express our creativity, and train our minds to plan and organise better.



However, playing video games for too long has its disadvantages, so we need to follow some healthy practices.

Healthy practices while playing video games:

1. Maintain good posture. Sit upright with a straight back and chin tilted slightly downwards.
2. It's easy to get caught up while playing a video game. But eyes need time to rest and look away from screen for a little while. Take frequent breaks and remember to blink while playing.
3. A video game is just a game. Don't take it personally and get angry while playing or if you lose. You can always replay the game.

HEALTHY PRACTICES FOR PLAYING VIDEO GAMES



Maintain a good posture.



Do not strain your eyes, blink frequently.



Schedule playtime outdoors and do not skip studies or other activities

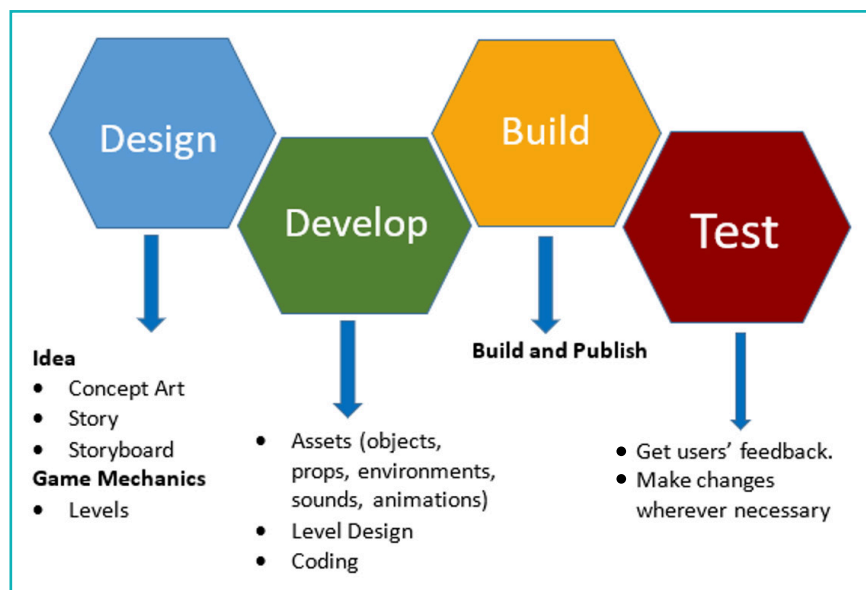


Do not get into fights while playing

Designing our own game has its own benefits. Apart from being able to create a game according to our needs, we develop some skills in the development process. Critical and analytical thinking, logical thinking and reasoning, creativity and time management are a few of these skills. We also develop soft skills such as storytelling abilities, effective communication and collaboration.

The game design process

The game design and development process involve the following stages:



1. **DESIGN:** In the design phase, we make a roadmap for game development. The idea for the game and its mechanics are described at this stage.
 2. **DEVELOP:** In the development phase, we create interactions between asset according to the game mechanics. Codes and scripts are created at this stage.
 3. **BUILD:** In the build phase, we build the game, export it and publish it on the decided upon platform.
 4. **TEST:** In the test phase, we ask a set of users to play the game and give feedback. Based on the feedback, we alter the game design or development.
- The first stage “Design” emphasises on conceptualisation. In this phase, we create a Game Design Document.

Game Design Document:

We document a game idea and its working in a **Game Design Document (GDD)**. This document plays an important role in creating a game. It helps the creator track game assets, how the game works and why they work in a certain way.

A GDD includes the following:

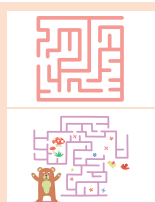
- a. Game overview:
 - Goal/purpose
 - Story
- b. Inspiration
- c. User interface
- d. Game characters and assets
- e. Level design
- f. Flowchart
- g. Game plan
- h. Abilities (if required - of the player or game)
- i. Sound (if used)

GAME OVERVIEW

Goal: Lorem ipsum dolor sit amet, consectetur adipiscing elit.
 Story: Lorem ipsum dolor sit amet, consectetur adipiscing elit, nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat. Ut wisi enim ad minim veniam, qui exerci tation ullamcorper suscipit lobortis nisl ut aliquip e commodo consequat. Duis autem vel eum iriure dolor in

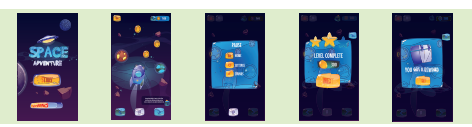


INSPIRATION




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USER INTERFACE



GAME CHARACTERS AND ASSETS

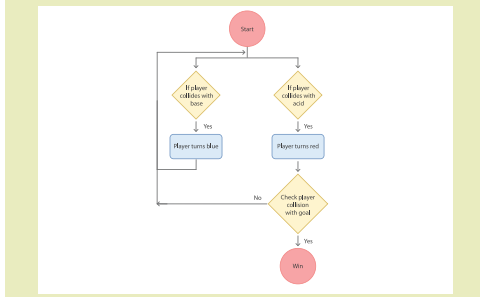


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FLOWCHART



GAME PLAN

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ABILITIES



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SOUND

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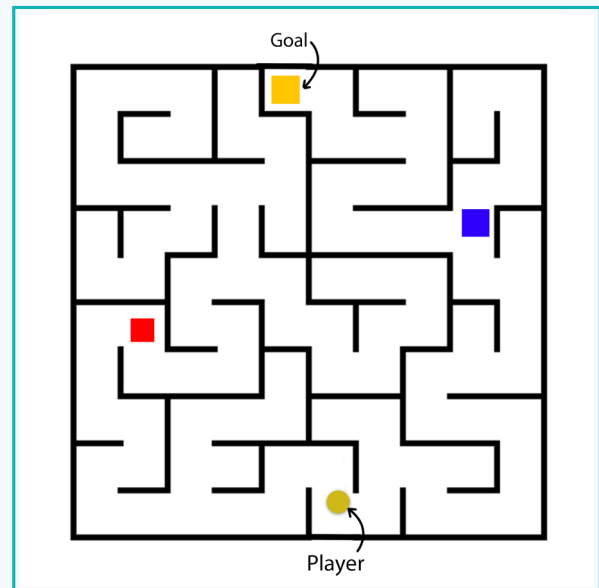
Activity

Create a Game Design Document (GDD) for a maze game by implementing the properties of a Litmus Test.

1. Game overview: The first stage includes an idea for the game and a game overview.

- A litmus test is used to identify whether a substance is acidic or alkaline. In this game, we will design a maze. The player will act as a litmus paper and change its colour when it touches an acidic or alkaline object. The player has to find a way out and also needs to fulfil a specific condition. When the player exits, it can win if its colour is red at the time.

2. User interface/Level design:



3. Game Characters/Assets for the game: Game assets include everything that goes into a game, including 2D or 3D models, sprites, and sound effects used by a game engine.

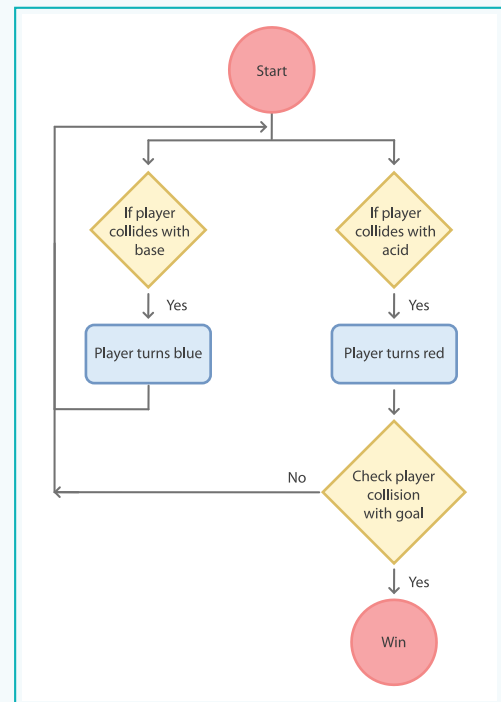
For our game, we require:

- A player
- Acid object
- Base object (Alkaline)
- Goal
- Colliders: Maze walls

Let's start by creating a flowchart.

Activity

4. Flowchart: A flowchart helps us plan and display the flow of the game and how each step is related. It also helps to analyse the process and check all possible inputs and outputs.



5. Game plan: The litmus game is a maze game where the player moves along the player area in search of the goal object. The walls of the maze serve as the obstacle for the player while trying to reach the goal.

- Game objects affect the player properties and change its colour.
- As the player touches the acidic object, the player turns red.
- As the player touches the alkaline object, the player turns blue.
- The goal of the player is to find and reach the goal object in acidic form.
- If the player reaches the goal in alkaline form, the win condition is not met. Then, the player needs to turn itself to acidic state and find the goal again.

Now that our GDD is ready, let us look at which platform can be used to build the game.

Game creation involves asset design and building a programme which specifies how the assets and game will work. A programme is built with a game engine.

Game Engines: A game engine is a software that provides game developers the features necessary to build a game. Different game engines serve different purposes.

In the following lessons, will explore a game engine called **Unity**.

Unity is a cross-platform game engine developed by Unity Technologies. The engine is used to create three-dimensional (3D) and two-dimensional (2D) games, as well as interactive simulations and other experiences. It has been adopted by industries outside video gaming too, such as film, automotive, architecture, engineering, and construction.



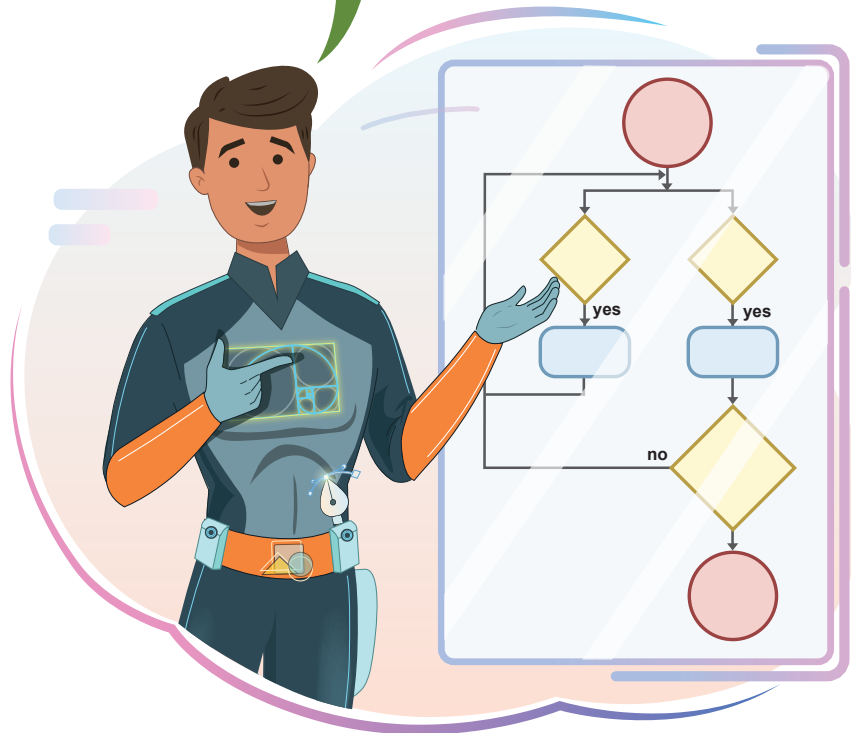
Exercise

Fill in the blanks

- 1 The game design process involves _____, _____, _____ and _____ stages.
- 2 A _____ helps us plan and display the flow of the game.
- 3 The documentation of a game idea and its working is called a _____.
- 4 The functionality of a game is built using _____.
- 5 The Unity game engine can be used to create _____ and _____ games, as well as interactive simulations and other experiences.

In this lesson, we learnt about video games and their effects on us, both positive and negative. We also learnt about the game design process which involves the stages Design, Develop, Build and Test. Different game engines help us create three-dimensional (3D) and two-dimensional (2D) games. Developing our own games is a fascinating process that we will explore soon!

A well-planned approach is necessary for any game development. It helps in logical sequencing of gameplay.



Tech Flash

- **Game design** : The process of planning and building a video game.
- **Game engine** : A software that provides game developers with the features necessary to build a game.